

CNS

By Dr.Riffat Ibrahim Riffat



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Parkinson's disease

- Parkinson's disease is a chronic neurodegenerative disease with an unknown cause.

- It is characterized by:

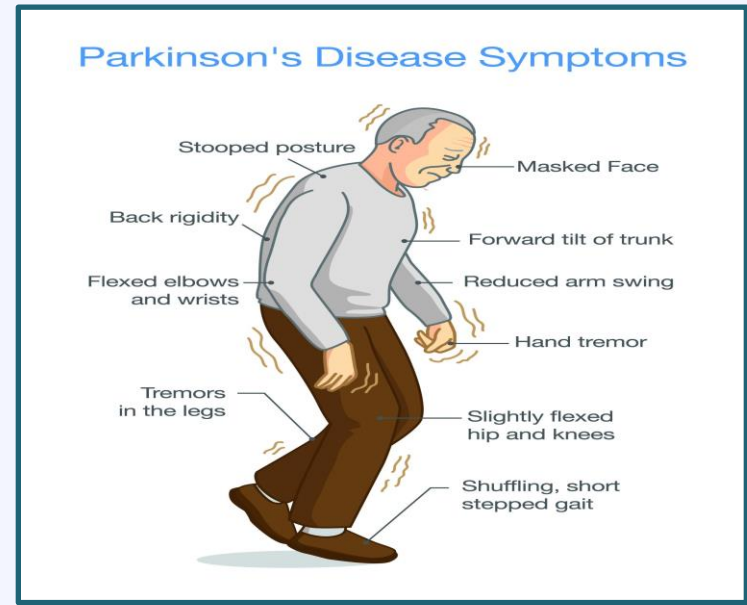
- Neurodegeneration of dopaminergic neurons (inhibitory) in the nigrostriatal pathway which causes acetylcholine (stimulatory) to take the upper hand.

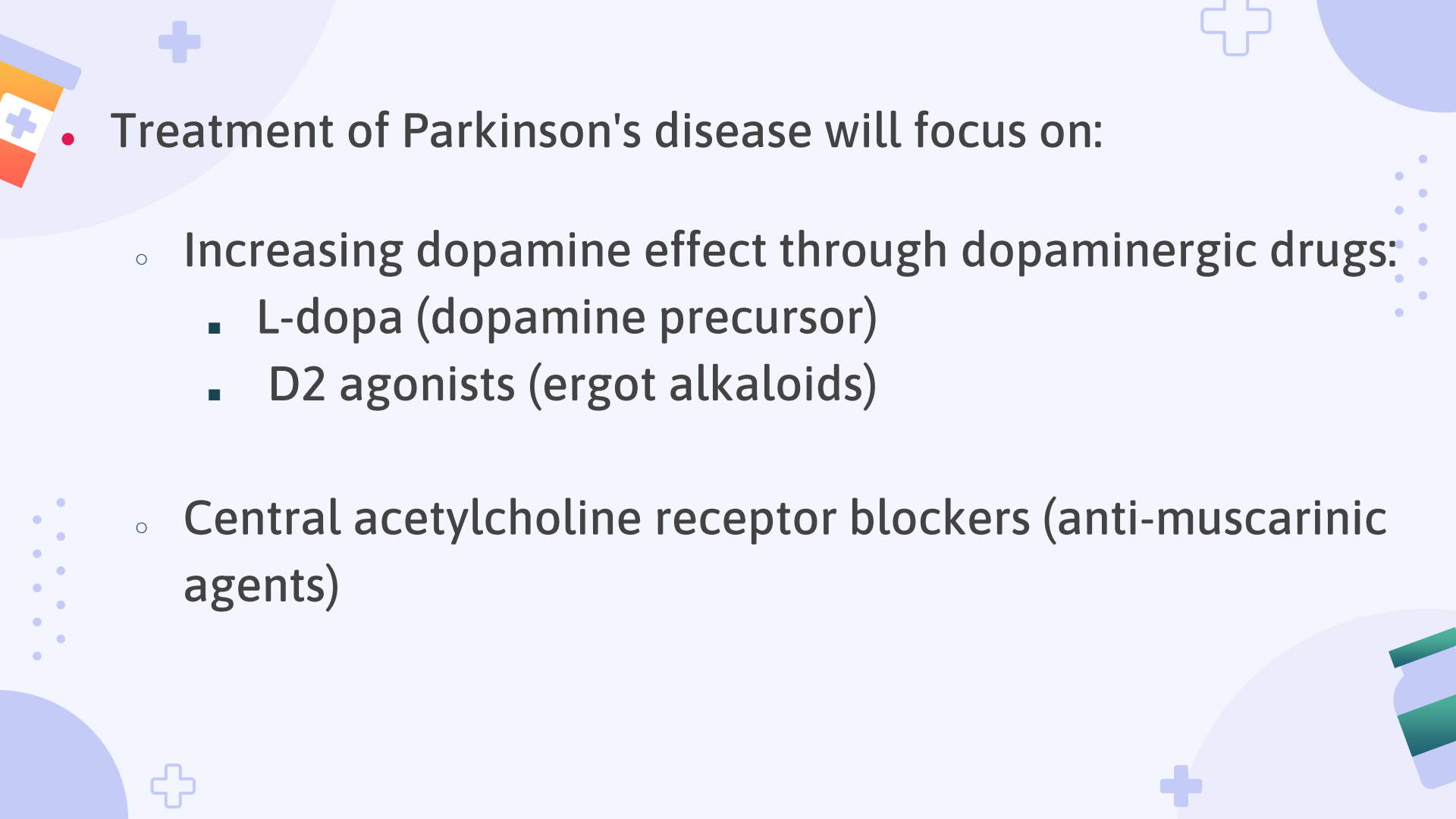
- Motor manifestations:

- Tremors.
- Muscle rigidity.
- Bradykinesia (shuffling gait).

- Non motor manifestations:

- Depression.
- Disturbance of sleep.
- Dementia



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- Treatment of Parkinson's disease will focus on:
 - Increasing dopamine effect through dopaminergic drugs:
 - L-dopa (dopamine precursor)
 - D2 agonists (ergot alkaloids)
 - Central acetylcholine receptor blockers (anti-muscarinic agents)

L-DOPA

Kinetics:

- It is a dopamine precursor as dopamine itself can not cross B.B.B.
- Well absorbed orally, but absorption is greatly affected by food.
- Very short half-life (1-2 hours) due to peripheral decarboxylation by hepatic decarboxylases and COMT metabolism resulting in:
 - the on/off phenomena, as most of the dose is metabolized peripherally not centrally which significantly reduces the amount that reaches CNS (actual target site).
 - Various peripheral side effects.
- Carbidopa is added to reduce peripheral decarboxylation of L-DOPA increasing the amount available to CNS from 5% of the dose to 60%

- Tolcapone or Entacapone (COMT inhibitors) are added to levodopa to prolong its duration of Action, as they prevent degradation of L-DOPA by COMT enzyme.
- COMT-inhibitors reduce On and Off phenomenon of levodopa/carbidopa and increase the dose available to CNS.
- Entacapone has replaced tolcapone due to tolcapone's hepatotoxicity.

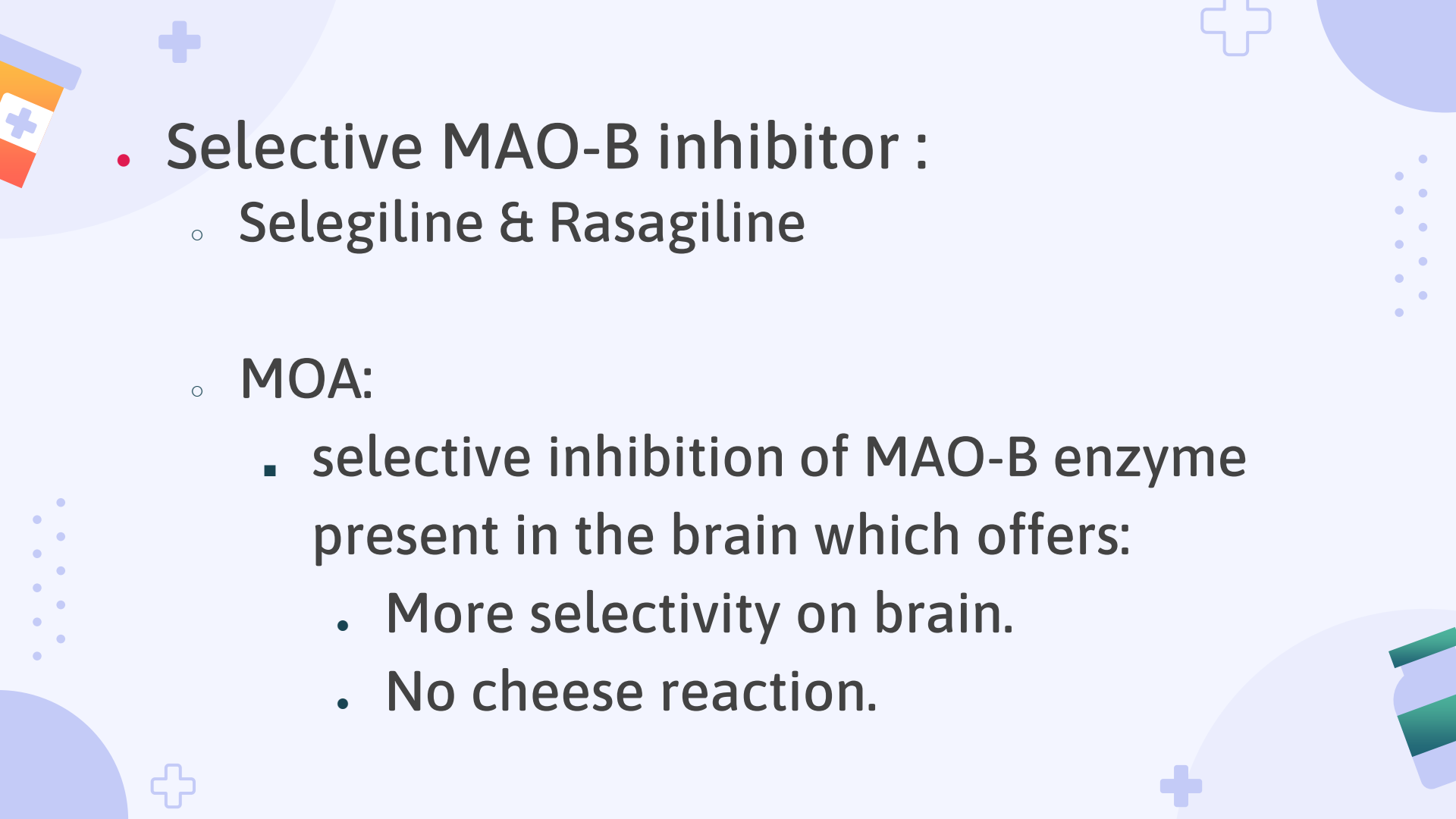
- Adverse effects:
 - CNS related side effects:
 - Nausea and Vomiting due to increased dopamine level in the vomiting center.
 - To solve this problem , use a Dopaminergic Blocker that can not pass BBB eg : (Domperidone).
 - Mood Changes (hallucination and delusion) due to increased dopamine level in mesocortical and mesolimbic pathways.
 - On and Off phenomenon (Sudden improvement followed by sudden immobility).

- **Peripheral side effects:**

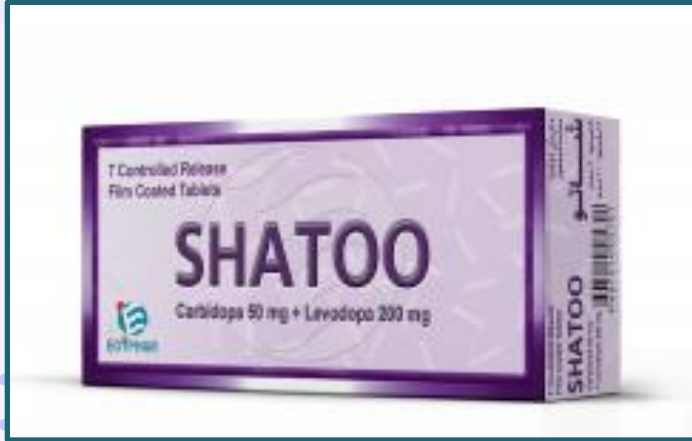
- Orthostatic hypotension and Arrhythmia due to:
 - Dopamine receptor activation in the heart (+ve Chrono and inotropic effect)
 - Dopamine receptor activation in the blood vessels (Vasodilatation)
- The most specific side effect of levodopa is brownish urine and sweat due to secretion of homovanillic acid (the metabolite of l-dopa) in urine and sweat.

- Important drug interactions:

- Levodopa should be taken on empty stomach to avoid leucine and Isoleucine interaction with levodopa absorption.
 - Pyridoxine(Vitamin B6) is contraindicated with Levodopa because of increased peripheral decarboxylation of levodopa.
 - Antipsychotic drugs (eg: Olanzapine) will prevent therapeutic effect of levodopa.
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- Selective MAO-B inhibitor :
 - Selegiline & Rasagiline
 - MOA:
 - selective inhibition of MAO-B enzyme present in the brain which offers:
 - More selectivity on brain.
 - No cheese reaction.

Brands



- Dopaminergic agonist:

- Ergot alkaloid : bromocriptin.

- MOA:

- Dopaminergic agonist.

- Bromocriptine has long Duration of action, It is not necessary to combine with COMT or MAO inhibitors. This explains why the drug exhibits no on and off phenomenon.

- Side effects:

- Orthostatic hypotension , Hallucination ,delusion , Nausea And vomiting (very similar to L-dopa due to it's agonistic action of dopamine receptors centrally and peripherally.
- Pulmonary fibrosis very specific to bromocriptin.



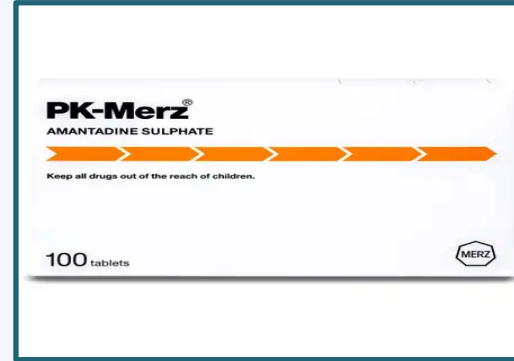
Amantadine: Antiviral drug that treats influenza and was by chance found to have anti parkinsonian effect.

- MOA:

- is unclear , but it may potentiate dopamine synthesis ,release and reuptake.

- Side effects:

- Nausea , vomiting ,dizziness and postural hypotension.
- levido reticularis(mottled skin due to dilation of capillaries).



- Anti-cholinergic drugs:
 - Benztropine ,Trihexyphenidyl ,Orphenadrine and Biperiden.
- MOA:
 - Blocking cholinergic transmission in the brain and correction the imbalance of Dopamine/Acetylcholine ratio. Resulting in an Improvement of motor manifestation.
- Adverse effects:
 - Anti-Cholinergic drugs worsen dementia , prostatic, hyperplasia, and glaucoma.
 - These drugs have the potential to be abused as in large doses they can cause hallucinations. (daturism)

Brands



Cases

- A 56-year-old man is diagnosed with Parkinson's disease.
- Carbidopa is prescribed. This agent reduces which of the following?
 - A. The activity of decarboxylase in the liver
 - B. The L -dopa dose necessary to achieve a therapeutic effect
 - C. The severity of L -dopa-associated dyskinesias
 - D. The time to onset of L -dopa's therapeutic effects

Answers

- Carbidopa, which does not penetrate the brain, reduces peripheral Dopa decarboxylase activity, and the metabolism of L -dopa. The therapeutic effect of L -dopa can be achieved at a lower dose than would be possible without carbidopa.

Cases

- Which one of the following combinations of antiparkinsonian drugs is an appropriate treatment plan?
 - A. Amantadine, carbidopa, and entacapone.
 - B. Levodopa, carbidopa, and entacapone.
 - C. Pramipexole, carbidopa, and entacapone.
 - D. Ropinirole, selegiline, and entacapone.
 - E. Ropinirole, carbidopa, and selegiline.

Answers

- Correct answer = B. To reduce the dose of levodopa and its peripheral side effects, the peripheral decarboxylase inhibitor, carbidopa, is coadministered. As a result of this combination, more levodopa is available for metabolism by catechol-O-methyltransferase (COMT) to 3-O-methyldopa, which competes with levodopa for the active transport processes into the CNS. By administering entacapone (an inhibitor of COMT), the competing product is not formed, and more levodopa enters the brain. The other choices are not appropriate, because neither peripheral decarboxylase nor COMT nor monoamine oxidase metabolizes amantadine or the direct-acting dopamine agonists, ropinirole and pramipexole.

Cases

- Peripheral adverse effects of levodopa, including nausea, hypotension, and cardiac arrhythmias, can be diminished by including which of the following drugs in the therapy?
 - A. Amantadine.
 - B. Ropinirole.
 - C. Carbidopa.
 - D. Tolcapone.
 - E. Pramipexole.

Answers

- Correct answer = C. Carbidopa inhibits the peripheral decarboxylation of levodopa to dopamine, thereby diminishing the gastrointestinal and cardiovascular side effects of levodopa. The other agents listed do not ameliorate adverse effects of levodopa.